Data lakes, warehouses, and lakehouses, in their standard implementation, rely heavily on a central data team. In our experience, this is their fundamental weakness when the company has a lot of data sources and complex usage patterns.

From a purely technical point of view, these three architectures may pose an excellent computational environment for a data mesh. This is because the elements of a data mesh may be constructed with existing infrastructure.

A data mesh will not require you to replace the technology into which your company has already invested millions. A data mesh will ensure that organizational inefficiencies of over-centralization do not throttle down your platform's potential.

The three previously described solutions have a common element—namely, the centralized data team. The next solution is on the other extreme of the centralization scale.

2.2.4 Data fabric

A *data fabric* is a technology-centric solution for data management. It provides a low-code or no-code platform to connect data sources with consumers alongside governance applied on top of it. This approach aims to simplify the integration and delivery of the data in a heterogeneous data environment across many platforms.

Data fabric solutions utilize AI/ML and knowledge graphs in addition to metadata to make an integration and delivery process as automated as possible. A data fabric is a vague concept, and, unfortunately, it is next to impossible to find a precise definition. As Gartner defines it (see http://mng.bz/yvjq and http://mng.bz/M5Gn), an ideal data fabric should have the components shown in figure 2.6.